

Erdős Problem #425

EP-425	L1 - Tractable	Partiellement résolu	Number Theory
--------	----------------	----------------------	---------------

Énoncé original (anglais, reproduction mot pour mot)

Let $F(n)$ be the maximum possible size of a subset $A \subseteqq \{1, \dots, N\}$ such that the products ab are distinct for all $a < b$. Is there a constant c such that $F(n) = \pi(n) + (c + o(1))n^{\frac{3}{4}}(\log n)^{-\frac{3}{2}}$? If $A \subseteqq \{1, \dots, n\}$ is such that all products $a_1 \cdots a_r$ are distinct for $a_1 < \dots < a_r$ then is it true that $|A| \leq \pi(n) + O(n^{\frac{r+1}{2r}})$?

Source et attribution. Énoncé reproduit depuis UnsolvedMath, jeu de données officiel sous licence CC BY 4.0.
Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Jeu de données: <https://huggingface.co/datasets/ulamai/UnsolvedMath>